
Deployment Object in Kubernetes (Rollout, Rollback, Versioning)

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Deployment manifest

```
root@ip-172-31-36-215: /home/ubuntu
kind: Deployment
apiVersion: apps/v1
metadata:
  name: mydeployments
spec:
  replicas: 2
  selector:
    matchLabels:
      name: deployment
  template:
    metadata:
      name: testpod
      labels:
        name: deployment
    spec:
      containers:
        - name: c00
          image: ubuntu
          command: ["/bin/bash", "-c", "while true; do echo Dexter Morgan; sleep 5; done"]
]
-- INSERT --
```

1.

(This manifest defines a Kubernetes Deployment resource. A Deployment is a higher-level abstraction for managing pods in Kubernetes, ensuring that the desired number of replicas of a pod are running and updated properly. mydeployments is the name of the deployment. This is how you will reference this deployment in the cluster. selector: Defines how the deployment identifies which pods to manage. A selector is used to match pods with specific labels. matchLabels: The deployment will look for pods with the label name: deployment to manage. This is a way for the deployment to target the pods it is responsible for. Deployment is a higher-level abstraction that provides a robust mechanism for managing applications, handling rolling updates and rollbacks, and ensuring availability. ReplicaSet focuses solely on maintaining the desired number of pod replicas without the features of automatic updates and rollbacks. Deployment: Manages the lifecycle of applications, including handling rolling updates and rollbacks when changes are made to the application. Deployments use ReplicaSets internally to manage the pods. ReplicaSet: Focuses purely on maintaining the number of replicas of a pod. It does not manage updates or rollbacks on its own.)

```
kind: Deployment
apiVersion: apps/v1
metadata:
  name: mydeployments
spec:
  replicas: 2
  selector:
    matchLabels:
      name: deployment
  template:
    metadata:
      name: testpod
    labels:
      name: deployment
    spec:
      containers:
        - name: c00
          image: ubuntu
          command: ["/bin/bash", "-c", "while true; do echo Dexter Morgan; sleep 5; done"]
```

How to list the deployment objects(i.e ReplicaSet)

```

root@ip-172-31-36-215:/home/ubuntu# vi mydeploy.yml
root@ip-172-31-36-215:/home/ubuntu# kubectl apply -f mydeploy.yml
deployment.apps/mydeployments created
root@ip-172-31-36-215:/home/ubuntu# kubectl get deployment
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
mydeployments        2/2     2             37s
root@ip-172-31-36-215:/home/ubuntu# kubectl get deploy
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
mydeployments        2/2     2             45s
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
mydeployments-79f5c9cbf5-c8t8l      1/1     Running   0           57s
mydeployments-79f5c9cbf5-pm9jk      1/1     Running   0           57s
root@ip-172-31-36-215:/home/ubuntu#

```

2.

commands:

kubectl get deploy

OR

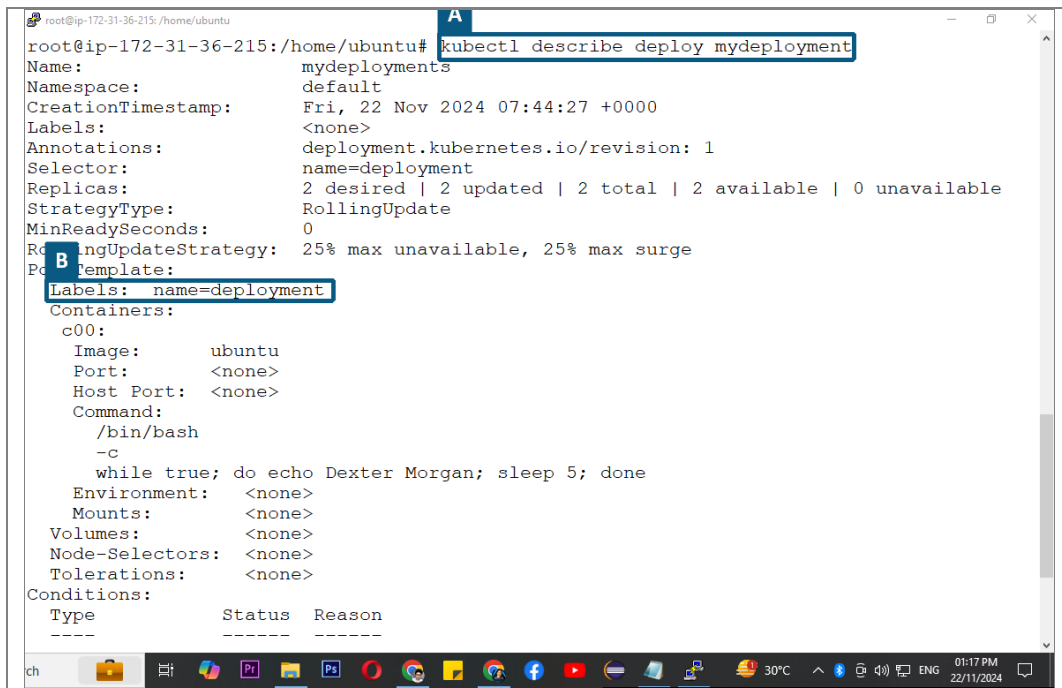
kubectl get deployment

NOTE:

Deployment: Supports rolling updates and rollbacks. When you update the image or configuration of the container, the Deployment ensures zero downtime by gradually updating pods.

ReplicaSet: Does not manage updates. If you want to update the pods, you would need to manually update the ReplicaSet, which won't handle graceful updates.

How to check deployment details



The screenshot shows a terminal window with the command `kubectl describe deploy mydeployment` executed. The output displays the details of the deployment, including its name, namespace, creation timestamp, labels, annotations, selector, replicas, strategy type, and update strategy. A red box highlights the command, and a red letter 'A' is next to it. Another red box highlights the 'Labels: name=deployment' line, with a red letter 'B' next to it.

```
root@ip-172-31-36-215:/home/ubuntu# kubectl describe deploy mydeployment
Name: mydeployments
Namespace: default
CreationTimestamp: Fri, 22 Nov 2024 07:44:27 +0000
Labels: <none>
Annotations: deployment.kubernetes.io/revision: 1
Selector: name=deployment
Replicas: 2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
PodTemplate:
  Labels: name=deployment
  Containers:
    c00:
      Image: ubuntu
      Port: <none>
      Host Port: <none>
      Command:
        /bin/bash
        -c
        while true; do echo Dexter Morgan; sleep 5; done
      Environment: <none>
      Mounts: <none>
      Volumes: <none>
      Node-Selectors: <none>
      Tolerations: <none>
  Conditions:
    Type          Status  Reason
    ----          -
    Type          Status  Reason
    ----          -
    Type          Status  Reason
    ----          -
```

3.

```
kubectl describe deploy deployment-object-name
kubectl describe deploy mydeployment
```

As per our instructions in the Yaml manifest Deployment object has created 2 replicas

```

root@ip-172-31-36-215:/home/ubuntu# kubectl get rs
NAME                                DESIRED    CURRENT    READY    AGE
mydeployments-79f5c9cbf5            2          2          2        4m52s
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME                                READY     STATUS    RESTARTS   AGE
mydeployments-79f5c9cbf5-c8t8l      1/1       Running   0           5m2s
mydeployments-79f5c9cbf5-pm9jk      1/1       Running   0           5m2s
root@ip-172-31-36-215:/home/ubuntu# kubectl delete pod ^C
root@ip-172-31-36-215:/home/ubuntu# kubectl delete pod mydeployments-79f5c9cbf5-c8t8l
pod "mydeployments-79f5c9cbf5-c8t8l" deleted
root@ip-172-31-36-215:/home/ubuntu# kubectl get rs
NAME                                DESIRED    CURRENT    READY    AGE
mydeployments-79f5c9cbf5            2          2          2        6m23s
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME                                READY     STATUS    RESTARTS   AGE
mydeployments-79f5c9cbf5-6n5ww      1/1       Running   0           40s
mydeployments-79f5c9cbf5-pm9jk      1/1       Running   0           6m26s
root@ip-172-31-36-215:/home/ubuntu#

```

4.

commands:

1. kubectl get rs
2. kubectl get pods
3. kubectl delete pod pod-name.

NOTE:

ReplicaSet will maintain the desired state(i.e replica=2) Even if we delete a pod it'll immediately create a new one

Changes occurred part-1

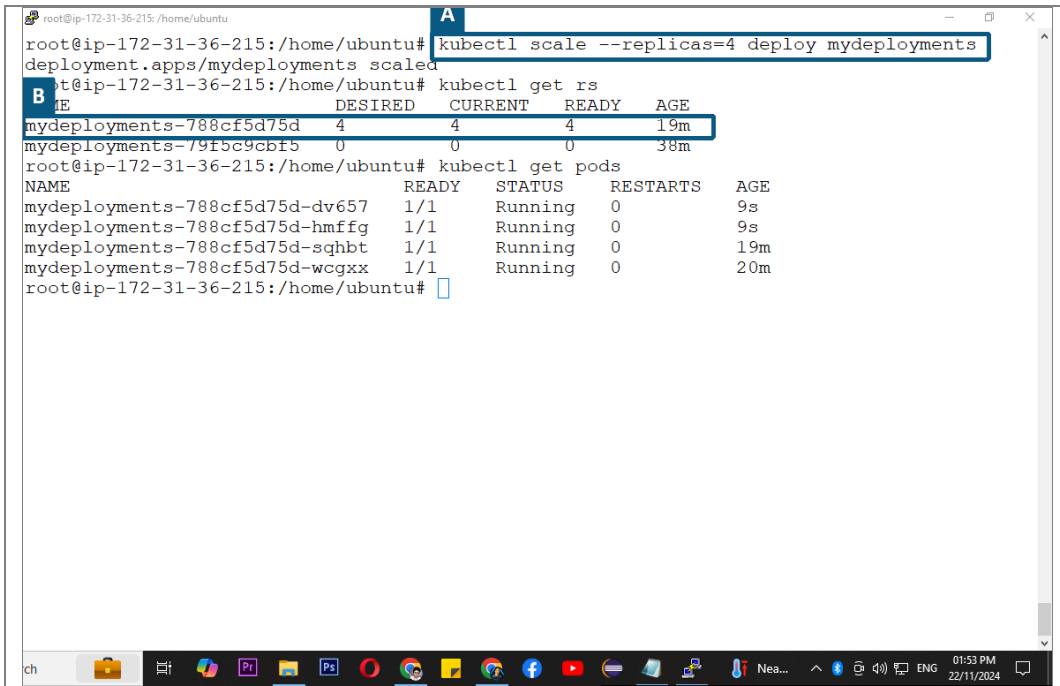
```
root@ip-172-31-36-215:/home/ubuntu# vi mydeploy.yml
root@ip-172-31-36-215:/home/ubuntu# kubectl apply -f mydeploy.yml
deployment.apps/mydeployments configured
root@ip-172-31-36-215:/home/ubuntu# kubectl get rs
A
E
DESIRED    CURRENT    READY    AGE
mydeployments-788cf5d75d    2          2        10s
mydeployments-79f5c9cbf5    0          0        18m
root@ip-172-31-36-215:/home/ubuntu#
```

8.

NOTE:

1. Previous replicas have been freezed and new Replicas are created
2. We can call it a new version

How to scale up Deployment objects(Replicas)



The terminal screenshot shows the following commands and output:

```
root@ip-172-31-36-215:/home/ubuntu# kubectl scale --replicas=4 deploy mydeployments
deployment.apps/mydeployments scaled
root@ip-172-31-36-215:/home/ubuntu# kubectl get rs

```

NAME	DESIRED	CURRENT	READY	AGE
mydeployments-788cf5d75d	4	4	4	19m
mydeployments-79f5c9c6f5	0	0	0	38m

```
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods

```

NAME	READY	STATUS	RESTARTS	AGE
mydeployments-788cf5d75d-dv657	1/1	Running	0	9s
mydeployments-788cf5d75d-hmffg	1/1	Running	0	9s
mydeployments-788cf5d75d-sqgbt	1/1	Running	0	19m
mydeployments-788cf5d75d-wcgxx	1/1	Running	0	20m

11.

```
kubectl scale replicas=*Desired number* deploy deployment-object-name
kubectl scale replicas=4 deploy mydeployments
```

How to rollout to previous version and check rollout history

```

deployment.apps/mydeployments scaled
root@ip-172-31-36-215:/home/ubuntu# kubectl get rs
NAME                                DESIRED    CURRENT    READY    AGE
mydeployments-788cf5d75d             4          4          4        19m
mydeployments-79f5c9cbf5             0          0          0        38m
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME                                READY      STATUS    RESTARTS   AGE
mydeployments-788cf5d75d-dv657      1/1       Running   0           9s
mydeployments-788cf5d75d-hmffg      1/1       Running   0           9s
mydeployments-788cf5d75d-sqhbt      1/1       Running   0          19m
mydeployments-788cf5d75d-wcgxx      1/1       Running   0          20m
root@ip-172-31-36-215:/home/ubuntu# kubectl rollout status deployment mydeployments
deployment "mydeployments" successfully rolled out
root@ip-172-31-36-215:/home/ubuntu# kubectl rollout history deployment mydeployments
deployment.apps/mydeployments
REVISION  CHANGE-CAUSE
1          <none>
2          <none>

root@ip-172-31-36-215:/home/ubuntu# kubectl get rs
NAME                                DESIRED    CURRENT    READY    AGE
mydeployments-788cf5d75d             4          4          4        23m
mydeployments-79f5c9cbf5             0          0          0        41m
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME                                READY      STATUS    RESTARTS   AGE
mydeployments-788cf5d75d-dv657      1/1       Running   0          3m35s
mydeployments-788cf5d75d-hmffg      1/1       Running   0          3m35s
mydeployments-788cf5d75d-sqhbt      1/1       Running   0          23m
mydeployments-788cf5d75d-wcgxx      1/1       Running   0          23m
root@ip-172-31-36-215:/home/ubuntu#

```

12.

commands:

1. `kubectl rollout status deployment deployment-object-name`
2. `kubectl rollout history deployment deployment-object-name`

NOTE:

- Two changes(versions) are reflecting as we have made changes twice.
- The replicas(pod) numbers will remain the same(i.e if we were using 4 replicas in the current operation even if we rollback the replica number will be the same even though we were working on 2 replicas in the previous version)
- However this command will not apply the rollout changes(previous version).

How to apply the rollout changes

```

root@ip-172-31-36-215:/home/ubuntu# kubectl rollout undo deploy/mydeployments
deployment.apps/mydeployments rolled back
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
mydeployments-788cf5d75d-dv657     1/1     Terminating    0          18m
mydeployments-788cf5d75d-hmffg     1/1     Terminating    0          18m
mydeployments-788cf5d75d-sqhbt     1/1     Terminating    0          38m
mydeployments-788cf5d75d-wcgxx     1/1     Terminating    0          38m
mydeployments-79f5c9cbf5-k5r54     1/1     Running         0          12s
mydeployments-79f5c9cbf5-pbxf5     1/1     Running         0          10s
mydeployments-79f5c9cbf5-xhr64     1/1     Running         0          16s
mydeployments-79f5c9cbf5-zqvwn     1/1     Running         0          16s
root@ip-172-31-36-215:/home/ubuntu# kubectl logs -f mydeployments-79f5c9cbf5-k5r54
Dexter Morgan
Dexter Morgan
Dexter Morgan
Dexter Morgan
Dexter Morgan
Dexter Morgan
Dexter Morgan
Dexter Morgan

```

13.

commands:

1. `kubectl rollout undo deploy/deployment-object-name`
2. `kubectl logs -f pod-name`

NOTE:

- The rollout changes have been applied previous pods are being terminated and new ones are being created with the previous code and configurations(such as os and other softwares).
- The echo command has been changes(rolled back to previous version).

Rollback changes

```
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
mydeployments-79f5c9cbf5-k5r54     1/1     Running   0           3m7s
mydeployments-79f5c9cbf5-pbxf5     1/1     Running   0           3m5s
mydeployments-79f5c9cbf5-xhr64     1/1     Running   0           3m11s
mydeployments-79f5c9cbf5-zqvwv     1/1     Running   0           3m11s
root@ip-172-31-36-215:/home/ubuntu# kubectl exec mydeployments-79f5c9cbf5-xhr64 -- cat /
PRETTY_NAME="Ubuntu 24.04.1 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.1 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
root@ip-172-31-36-215:/home/ubuntu#
```

14.

As we can see the os has also been rolled out to the previous version i.e “ubuntu”

How to DELETE the Deployment object

```
root@ip-172-31-36-215:/home/ubuntu# ls
kubectl  manifest1.yml  manifest2.yml  manifest3.yml  manifest4.yml  manifest5.yml  myreplica.yml  myrs.yml
labels.yml  manifest2.yml  manifest3.yml  manifest4.yml  mydeploy.yml  myreplicaset.yml  newmanifest.y
root@ip-172-31-36-215:/home/ubuntu# kubectl delete -f mydeploy.yml
deployment.apps "mydeployments" deleted
root@ip-172-31-36-215:/home/ubuntu# kubectl get rs
No resources found in default namespace.
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
mydeployments-79f5c9cbf5-k5r54     1/1     Terminating   0           6m29s
mydeployments-79f5c9cbf5-pbxf5     1/1     Terminating   0           6m27s
mydeployments-79f5c9cbf5-xhr64     1/1     Terminating   0           6m33s
mydeployments-79f5c9cbf5-zqvwv     1/1     Terminating   0           6m33s
root@ip-172-31-36-215:/home/ubuntu#
```

15.

`kubectl delete -f mydeploy.yml`